Revolutionizing Computing: Harnessing Mechanical Energy from Typing for Sustainable Power Generation

Sienna Giuseppi
Pelham Memorial High School
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Abstract

To resist detrimental and irreversible climate damage, our dependency must shift from fossil fuel-derived energy to renewable energy sources. Unfortunately, current popular renewable energy sources require the construction of massive structures that distress their environments (ex. wind turbines, and nuclear power plants). Electricity production is one of the greatest contributors to CO_2 emissions, with its utilizations ranging from the heating of households to the charging of electronic devices. Piezoelectricity is the phenomenon in which electrical charge is accumulated in response to applied mechanical stress. With laptops presenting themselves as one of the most widely-used electronic devices in the world, and mechanical energy constantly being dispersed onto the keys whenever in use, it was hypothesized that piezoelectric material can charge the device by harnessing the mechanical energy from each keystroke to power the device. A series of meticulous experiments and tests, including the measurement of average force during typing, underscore the feasibility of piezoelectric technology in energy harvesting. The experiments reveal that each piezoelectric material responds with positive voltage generation when subjected to the forces typically applied during typing, providing substantial evidence for the viability of this approach. The research culminates in a tangible computer key design that integrates a piezoelectric element, (APC Item 1738), to capture the mechanical energy of typing. The results are promising, with each keystroke producing a maximum voltage reading of 0.403V when applying a force of 0.6 N. While this figure may seem modest in isolation, it signifies the potential to accumulate energy over time, thus supporting the concept of creating self-sustaining computers powered by renewable energy from typing.

Introduction

Global greenhouse gas (GHG) emissions exhibit a persistent upward trajectory, and the imperative to address the escalating climate crisis intensifies. The Paris Climate Accord's explicit ambition of constraining global warming to below 2 degrees Celsius, with a preference of 1.5 degrees Celsius, signifies a collective commitment to mitigating climate-related risks and safeguarding ecological systems and human well-being. Yet, the current trend of emissions reduction endeavors indicates a substantial shortfall in achieving the requisite decarbonization rates (Ivanova et al, 2020). The energy sector is a significant contributor to carbon emissions. In its 2022 CO_2 Emission report the International Energy Agency (IEA) shared that Energy-related CO_2 emissions reached an all-time high of 36.8 Gt, increasing by 0.9% over the year. It was also reported that global electricity demand increased by 2.7%, resulting in the power sector (electricity and heat generation) experiencing the largest sectoral increase and a record-high emission rate of 14.6 Gt of CO_2 (IEA). These increases in emissions lead to rising global temperatures, extreme weather events, and loss of global biodiversity. Additionally, the need to transition away from fossil fuels is underscored by the projected depletion of most fossil fuel reserves in the coming decades, as based on current rates of consumption, coal will be the only remaining fossil fuel after 2047 (Shafiee and Topal, 2009).

Exploration of alternative energy sources and the transition towards greater reliance on renewables is essential for the preservation and enhancement of our planet's condition for generations to come. Our surroundings offer a diverse array of energy sources, encompassing wind, sunlight, heat, and mechanical vibrations, all of which possess the potential for generating clean energy via energy harvesting mechanisms and thus contribute to the conservation of finite, nonrenewable resources (Akinaga 2020). Nevertheless, practical constraints are inherent in several well-established renewable sources, necessitating substantial infrastructure deployments (e.g., solar cells, geothermal power plants, nuclear power plants, tidal turbines, and wind turbines) (Kumar et al. 2013). While these conventional energy generation methods have proven effective in harnessing clean energy, the imperative for renewable energy sources to expand remains crucial to surpass carbon-intensive energy production processes. Methods used to generate electrical power from ambient or waste energy sources also include photovoltaic, thermoelectric, piezoelectric, pyroelectric, radio frequency (RF), electromagnetic induction, electrostatic, and capacitive methods (Edwards and Gould, 2016). Their contemporary focus has been placed on designing self-powered systems that can fully remove the need for, or at least supplement, the electrical power supplied by onboard batteries or other power sources, thus alleviating the masses of carbon-related energy production in everyday devices. The prerequisite to popularizing and installing clean energy into smaller aspects of everyday life is a call for more widespread implementation of renewable energy sources.

Piezoelectricity

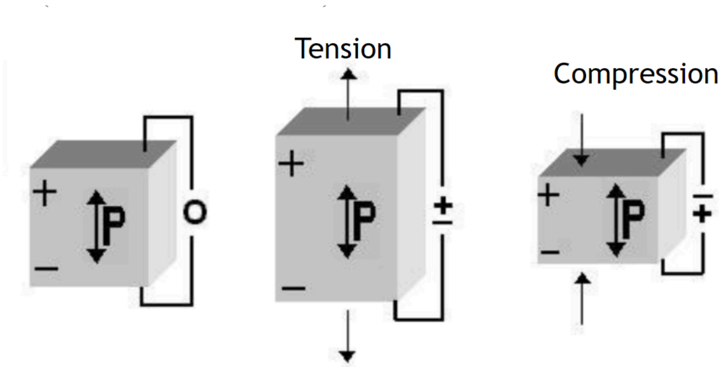
Mechanical energy exists ubiquitously, emanating from sources such as vibrations, vehicle operations, and human physical activities (Sun et al., 2019). However, in most scenarios, this energy remains overlooked. The historically limiting factor in the widespread adoption of ambient energy sources has been the relatively low levels of electrical power and efficiencies generated by these techniques, however, in applications where the energy that could be generated is not utilized and effectively wasted, these limitations do not preclude their implementation (Edwards and Gould, 2016).

Piezoelectric materials possess the capacity to undergo deformation when subjected to an electric field, a phenomenon known as the reverse piezoelectric effect. This property makes them valuable in an array of technologies, including accelerometers, sensors, actuators, transducers, and acoustic devices. As a result, these materials play a pivotal role in advancing diverse fields, including ultrasound imaging, medical diagnostics, energy harvesting, and vibration sensing (American Piezo, 2023; Sezer and Koc, 2021). Furthermore, the direct piezoelectric effect, (as pictured in Figure 1) involves the accumulation of electrical charge in specific solids in response to applied mechanical stress and garners considerable interest due to its high power density and compatibility with integration into various host structures (Sharma et al., 2019; Yan et al., 2021).

There are three primary approaches for harnessing electrical energy from vibrations as a source of kinetic energy: electromagnetic, electrostatic, and piezoelectric methods (Akinaga, 2020). Piezoelectric energy harvesting technology exploits the electromechanical coupling characteristics of piezoelectric materials to convert mechanical energy into electrical energy directly. The quantity of charge generated is contingent upon factors such as force, material properties, and the electromechanical configuration of the piezoelectric structure (Sharma et al., 2019). Piezoelectric power generation offers advantages such as independence from external power sources and compact size compared to electromagnetic and electrostatic generation (Xu et al., 2017). In light of the prevailing energy crisis and concerns about global warming, piezoelectric energy harvesting technologies have been integrated into piezoelectric energy harvesters, designed to capture clean energy, which has only gained substantial attention in the past decade.

Figure 1.

The direct piezoelectric effect (Kumar et al. 2013)



PZT Ceramics

The discovery of the piezoelectric effect (Fig.1) is credited to Brother's Curie in the 1880s. Recent research in the last decade has led to the development of piezoelectric ceramic materials such as lead zirconate titanate (PZT), lead magnesium niobate-lead titanate (PMN-PT), barium titanate (BT), and potassium sodium niobate (KNN). Due to their excellent piezoelectric properties, piezoelectric ceramics have been widely explored in power generation.

Incorporating minor quantities of a donor dopant into a ceramic formulation induces the formation of metal (cation) vacancies within the crystal arrangement. This alteration enhances the influence of extrinsic elements on the piezoelectric attributes of the ceramic material. The outcomes of such formulations, known as soft ceramics, are characterized by notable electromechanical coupling coefficients (k), significant piezoelectric constants, elevated permittivity, substantial dielectric constants ($\mathbf{K^T}$), pronounced dielectric losses **(tan δ)**, diminished mechanical quality factors ($\mathbf{Q_m}$), and suboptimal linearity. In comparison to hard ceramics, soft ceramics generate more extensive displacements and broader signal bandwidths, but they display increased hysteresis and heightened susceptibility to depolarization and deterioration. The relatively lower Curie points **((°C)**)** (typically below 300°C) of soft ceramics necessitate their utilization at lower temperatures. Their considerable permittivity and dielectric dissipation factor values usually limit or preclude the application of soft ceramics in scenarios demanding concurrent high-frequency inputs and high electric fields. Consequently, soft ceramics find primary utility in sensing applications rather than power-related applications.

Alternatively, incorporating an acceptor dopant within a ceramic formulation triggers the generation of oxygen (anion) vacancies within the crystal structure. In contrast to soft ceramics, hard ceramics exhibit characteristics largely contrary to them. This includes Curie points surpassing 300°C, modest piezoelectric charge constants, substantial electromechanical coupling coefficients, and considerable mechanical quality factors. Hard ceramics are also less prone to polarization or depolarization. While generally more stable than their soft counterparts, hard ceramics are unable to achieve comparable extensive displacements. They are, however, compatible with high mechanical loads and elevated voltages. (American Piezo, 2023)

Due to their versatility, piezoelectric ceramics are also made in a variety of sizes and shapes that can affect their voltage output.

The following equation defines piezoelectric energy harvesting figures of merit:

$$\mathrm{FoM}_{ij} = \frac{d_{ij}^{2}}{\varepsilon_0 \varepsilon_{33}^{T}}$$

This equation represents the piezoelectric energy harvesting figure of merit under mechanical stress, represented by FoMij. dij is the piezoelectric charge coefficient, ε0 is the permittivity of free space (constant of 8.85418782 × 10-12 m-3 kg-1 s4 A2), and εT33 is the relative permittivity at constant stress (Yan et al. 2021, Preimesberger et al. 2020). According to this equation, the piezoelectric charge coefficient, dij, and the relative permittivity, εT33, are highly influential to the harvested energy density for applied stress.

While piezoelectric ceramics do express high dij coefficients, their εT33 is also high, resulting in a relatively low piezoelectric figure of merit. (Yan et al. 2021). Piezoelectric ceramics are brittle and dense, which is a current drawback to their applications in forthcoming energy harvesting devices. Thus many studies have begun to focus on more flexible piezoelectric material.

Not all ceramics have identical qualities. Table 1 is taken from APC International, a leading distributor of piezoelectric material for a variety of applications. Though all of their materials are based on Lead Zirconate Titanate (PZT), slight differences in manufacturing processes yield hundreds of different materials with diverse specifications.

Table 1.

Piezoelectric properties according to APC International

Note: APC Materials 840, and 850 were used in this study

APC Material:	840	841	850	854	855	860	880
Navy Equivalent	Navy I	Hybrid	Navy II	Navy V	Navy VI	Porous	Navy III
Relative Dielectric Constant							
K^T	1275	1375	1900	2750	3300	1200	1050
Dielectric Dissipation Factor (Dielectric Loss(%))*							
tan δ	0.60	0.40	≤ 2.00	≤ 2.00	≤ 2.50	≤ 2.00	0.40
Curie Point (°C)**							
T_c	325	320	360	250	200	360	310
Electromechanical Coupling Factor							
k_p	0.59	0.60	0.63	0.66	0.68	0.50	0.50
k_{33}	0.72	0.68	0.72	0.68	0.76	0.45	0.62
k_{31}	0.35	0.33	0.36	-	0.40	-	0.30
k_{15}	0.70	0.67	0.68	-	0.66	-	0.55
d_{33}	290	300	400	600	630	380	215
$-d_{31}$	125	109	175	260	276	-	95
d_{15}	480	450	590	625	720	-	330
Piezoelectric Voltage Constant (10^{-3} Vm/N or 10^{-3} m^2/C)							
g_{33}	26.5	25.5	24.8	25.5	21.0	38.0	25.0
$-g_{31}$	11.0	10.5	12.4	-	9.0	-	10.0
g_{15}	38.0	35.0	36.0	-	27.0	-	28.0
Young's Modulus (10^{10} N/m^2)							
Y^E_{11}	8.0	7.6	6.3	6.0	5.9	-	9.0
Y^E_{33}	6.8	6.3	5.4	5.2	5.1	-	7.2
Frequency Constants (Hz*m or m/s)							
N_L (longitudinal)	1524	1700	1500	-	1390	-	1725
N_T (thickness)	2005	2005	2040	2000	2079	1390	2110
N_P (planar)	2130	2055	1980	1972	1920	1900	2120
Density (g/cm^3)							
ρ	7.6	7.6	7.6	7.6	7.6	6.6	7.6
Mechanical Quality Factor							
Q_m	500	1400	80	70	65	50	1000
Acoustic Impedance (Mrayl)							
Z_a	-	-	31.5	-	-	16.5	-

The values listed above pertain to test specimins. They are for reference purposes only and cannot be applied unconditionally to other shapes and dimensions. In practice, piezoelectric materials show varying values depending on their thickness, actual shape, surface finish, shaping processes and post-processing.

Overall, piezoelectric elements in any form can convert strain energy into electrical energy, which could then be adapted for any low-power electronic devices (Yan et al. 2021). Collecting residual energy already exerted onto the device to convert into usable electrical energy is a positive and practical implication of piezoelectricity (Urroz-Montoya1 et al. 2019).

Computer Application

In 2016, the American Community Survey revealed that 89% of American households owned at least one computer, highlighting the ubiquitous presence of computers in our daily lives (census.gov). The routine use of computers translates to an annual average energy consumption of 150 kilowatts, primarily driven by the necessity to charge these devices regularly. Most household electricity is derived from nonrenewable energy sources, meaning device charging directly amplifies Consequently, the perpetual need for electronic device charging has become a common issue, often exacerbated by the frustration of forgetting a charger or lacking access to a convenient charging outlet, leading to a decline in productivity. The appeal of laptops, as opposed to desktop computers, lies in their portability; however, this freedom is curtailed when these devices remain tethered to an electrical outlet for extended periods.

Gap in Knowledge + Purpose

The innovative facet of piezoelectric technology is its capacity to harness energy from the immediate environment of a device, offering an enticing solution for on-the-go device charging. An average clerical worker registers more than 1700 keystrokes per hour, maintaining this pace for approximately 8 hours a day over several workdays, as documented by the Beijing University of Technology in 2012. Each keystroke on a keyboard yields a quantifiable amount of mechanical energy. When these actions are extrapolated to mirror the work patterns of an average office worker, a substantial reservoir of usable mechanical energy materializes. This observation suggests the feasibility of implementing a piezoelectric solution to enable self-charging capabilities in electronic devices.

Piezoelectricity holds the potential to revolutionize the autonomy of laptop charging, endorsing cleaner energy practices and alleviating the need for frequent, restrictive charging, as demonstrated by Yan et al. in 2021. Leveraging renewable energy sources from the immediate surroundings of the device offers a procurable power source, negating the need for daily reliance on a nearby power outlet. The overarching objective of this dual-phase research project is to establish a proof of concept for a keyboard system that harnesses mechanical energy generated during typing, thereby transforming it into usable electricity through the integration of piezoelectric materials.

Objectives + Hypotheses

Phase I

1. Explore the use of piezoelectric elements as an energy harvester
2. Determine the reliability of piezoelectric transducers as a power source for a laptop

H_1. Faster typing speed will yield a greater amount of harvested energy

Phase II

1. Compare the qualities of piezoelectric soft and hard ceramics
2. Explore the magnitude of forces on a computer to determine piezoelectric feasibility
4. Design and Build a computer key that harvests mechanical energy
3. Test piezoelectric Responses

Design and Engineering

Over the past four years, I have investigated piezoelectric energy harvesting technology and its applications. My initial idea for this project was inspired during my freshman year when I was tasked with coming up with a hypothetical research plan in a field of my interest. I took an interest in the topic of climate change and decided to focus my research plan on something that would help lower an individual's carbon footprint in daily life. I read multiple articles suggesting and testing piezoelectric feasibility in floor tiles, which showed promise in harnessing energy from mechanical stress, such as footsteps. I then made the connection between the mechanical energy used to type on a laptop, and the sum of laptops that are used and charged each day worldwide. Finding that most electricity is not produced renewably, I set a goal to modify a laptop so that the mechanical energy exerted while typing can be used to charge the laptop, thus reducing massive amounts of non-renewable sourced electricity while also taking a strain off of electrical grids by creating a self-sustaining device. This paper represents the culmination of my research efforts over two years, serving as the second iteration of the design prototype.

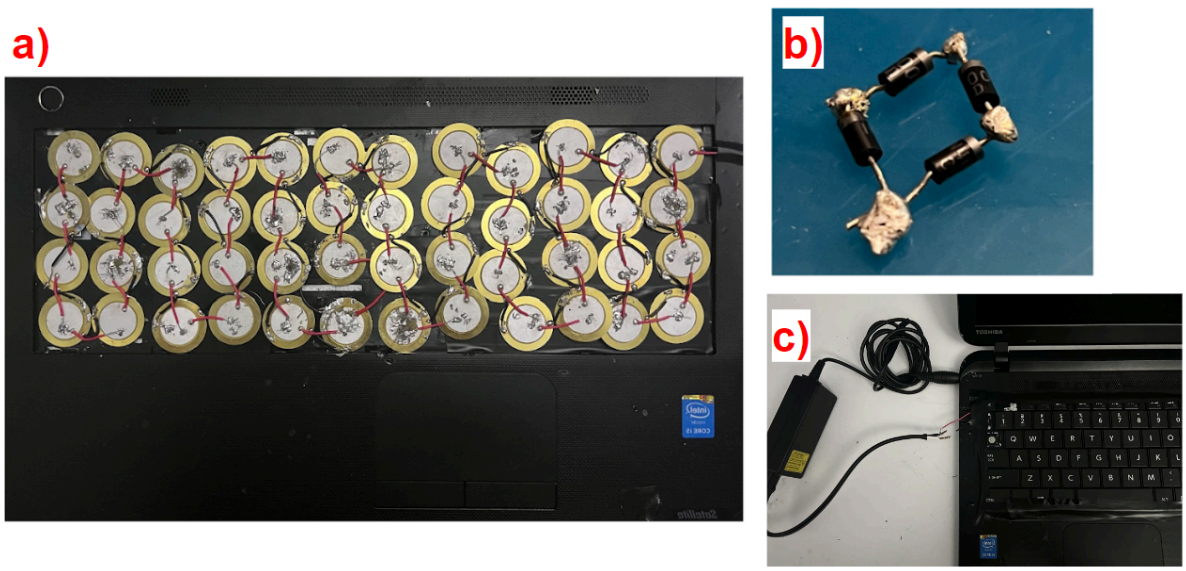
Phase I- Pilot Study

By 2022 I had read 36 journal articles, and independently designed and built my first prototype. The methodology involved designing a piezoelectric keyboard, utilizing a 2014 Toshiba Satellite laptop as the base for the project, and independently overseeing all modifications to the computer. The objective was to harness mechanical energy from keystrokes and convert it into electrical energy efficiently. Safety measures were taken during assembly, including the removal of the laptop's battery and covering exposed metal areas with electrical tape. The keyboard was removed from the laptop, and piezo disk transducers with a 35mm diameter were positioned beneath the keyboard to capture vibrations produced during typing. The electrical energy generated was then channeled back into the laptop through repurposed portions of the computer charger.

A chain of 48 piezoelectric transducers was aligned beneath the keyboard, and soldered in parallel to increase amperage. Digital multimeters were used to ensure that all piezos were correctly producing voltage, and the keyboard was reattached. To avoid damaging the battery of the computer through constant charging, the charger was adapted allowing users to choose when to connect the laptop to the piezoelectric power source. A bridge rectifier was added to convert AC energy from the piezoelectric transducers into usable DC energy for the computer. Statistical analyses were conducted, examining piezoelectric data in relation to typing speed (determined by an online typing test) and the voltage generated during typing. Data collection was facilitated by the Vernier GoLink USB sensor interface and Vernier Graphical Analysis software.

Figure 2

Phase I prototype components



Note: **a)** Piezoelectric elements chained together underneath the keyboard; **b)** diode bridge rectifier; **c)** charging cable attached to the piezoelectric chain.

Phase II

I was independently responsible for designing and building the piezoelectric computer key, including determining and gathering the appropriate materials. The key prototype was designed using Shapr3D and Makerbot software, and the piezoelectric ceramics were purchased from American Piezo. I came up with suitable methods to test my design's ability to harness and convert mechanical energy into electrical energy and performed the test on my own. I was also responsible for constructing the methods of testing and then analyzing the results.

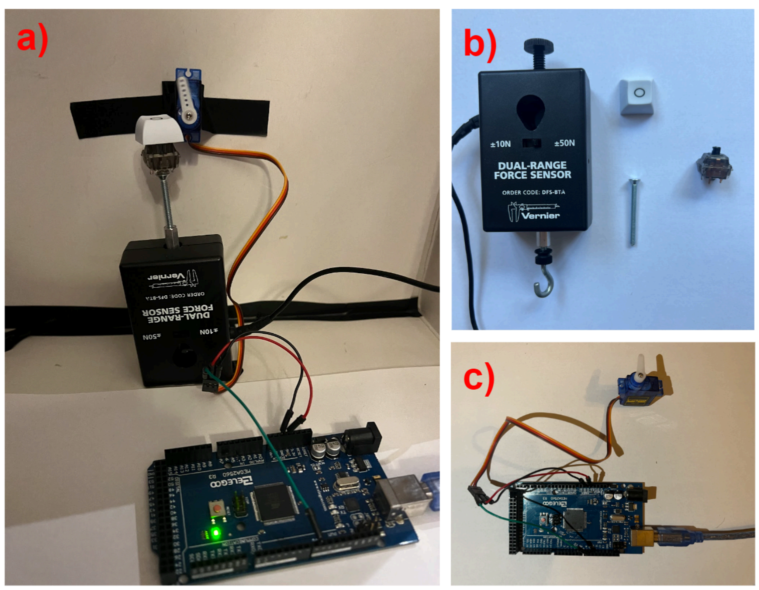
Key Pressure

To ensure the utmost accuracy of my design and testing procedures in replicating the typing process on a laptop, I undertook the acquisition of an average force curve for a single keystroke. Employing a mechanical key switch, I affixed it to a screw, which was then coupled with a Vernier Force Probe (Figure 3b). The ensuing experimental protocol involved the act of tapping the key, simulating the act of typing on a computer keyboard, to elucidate pertinent details of keystroke mechanics.

I then constructed and coded a circuit using an SG90 micro servo to rotate 180°(Figure 3c). This mechanism was then assembled and coded to strike a key with a determined force that matches the force curve of a typical keystroke. The purpose of this was to reduce human error and maintain consistency in force across all tests.

Figure 3

Pressure test



Note: **a)** SG90 motor circut and pressure probe setup; **b)** Vernier pressure probe, key cap and key switch, and screw; **c)** SG90 motor circuit.

Material Selection

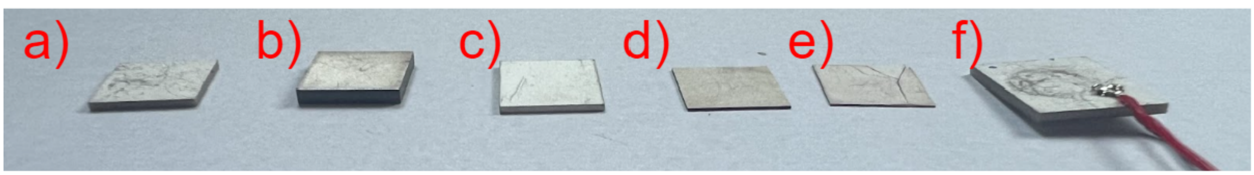
My selection of piezoelectric ceramic material for this study hinged on the utilization of APC International's PZT (Lead zirconate titanate) ceramics. The choice of PZT ceramics was motivated by extensive background research in the field and its wide range of current applications, rendering it the most economically viable energy medium for the prototype at hand. The specific PZT used in this configuration could be substituted to incorporate the most recent material developments, as the primary objective was to determine the overall viability of PZT material in the energy generation of the device. This investigation also aimed to discern the distinctions between hard and soft piezoelectric materials as energy harvesters

Six distinct materials were selected (Figure 4), each characterized by variations in dimensions, encompassing surface area and thickness. Furthermore, materials were chosen for their divergent properties, encompassing electromechanical coupling factors, and piezoelectric constants, among others, to detect the parameters that exert the most substantial influence on the study's outcomes. Throughout these selections, the electrode pattern remained constant as solid, as did the shape, maintained as a plate. The general size constraints also adhered to the dimensions suitable for accommodating a standard computer key.

To choose which material I used in the computer key, I soldered a live and neutral wire to each piezoelectric element and stressed each material by the average force of a keystroke. The voltage output was recorded and graphed with a Vernier Voltage Probe.

Figure 4

Piezoelectric Material (PZT)



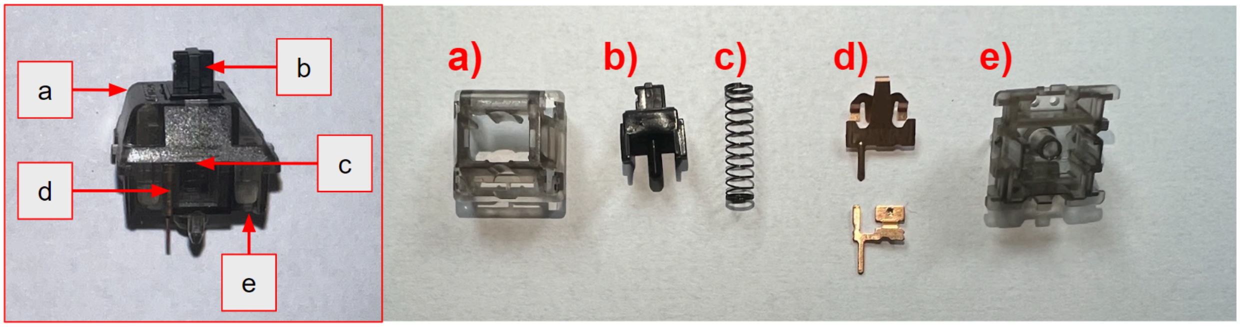
Note: **a)** Item 1738: *Material 850; Dimensions 10mm x 10mm; Thickness 0.80mm;* **b)**Item 248: *Material: 840; Dimensions 10.00mm x 10.00mm, Thickness 1.50 mm;* **c)** Item 1726: *Material: 850; Dimensions 11.6mm x 10.10 mm, Thickness 0.70 mm;* ***d)*** Item 1687: *Material: 840; Dimensions 10.00 mm x 10.00 mm, Thickness 0.28 mm;* **e)** Item 1377: *Material: 850; Dimensions 10.00mm x 10.00 mm, Thickness 0.19 mm;* **f)** Item 1365: *Material: 850; Dimensions: 14.00 mm x 12.00 mm, Thickness: 1.00 mm;*

Key Design

In my original design prototype, comprehensive consideration was given to the entire keyboard as a unit, introducing substantial variability into the data collection process. Subsequently, my new design focused efforts on the construction of an individual computer key, with the intention of facilitating reproducibility across the entire keyboard.

Figure 5

Key Switch components



Note: **a)** Upper Housing; **b)** Stem; **c)** Spring; **d)** Metal Contact Leaves; **e)** Housing Base

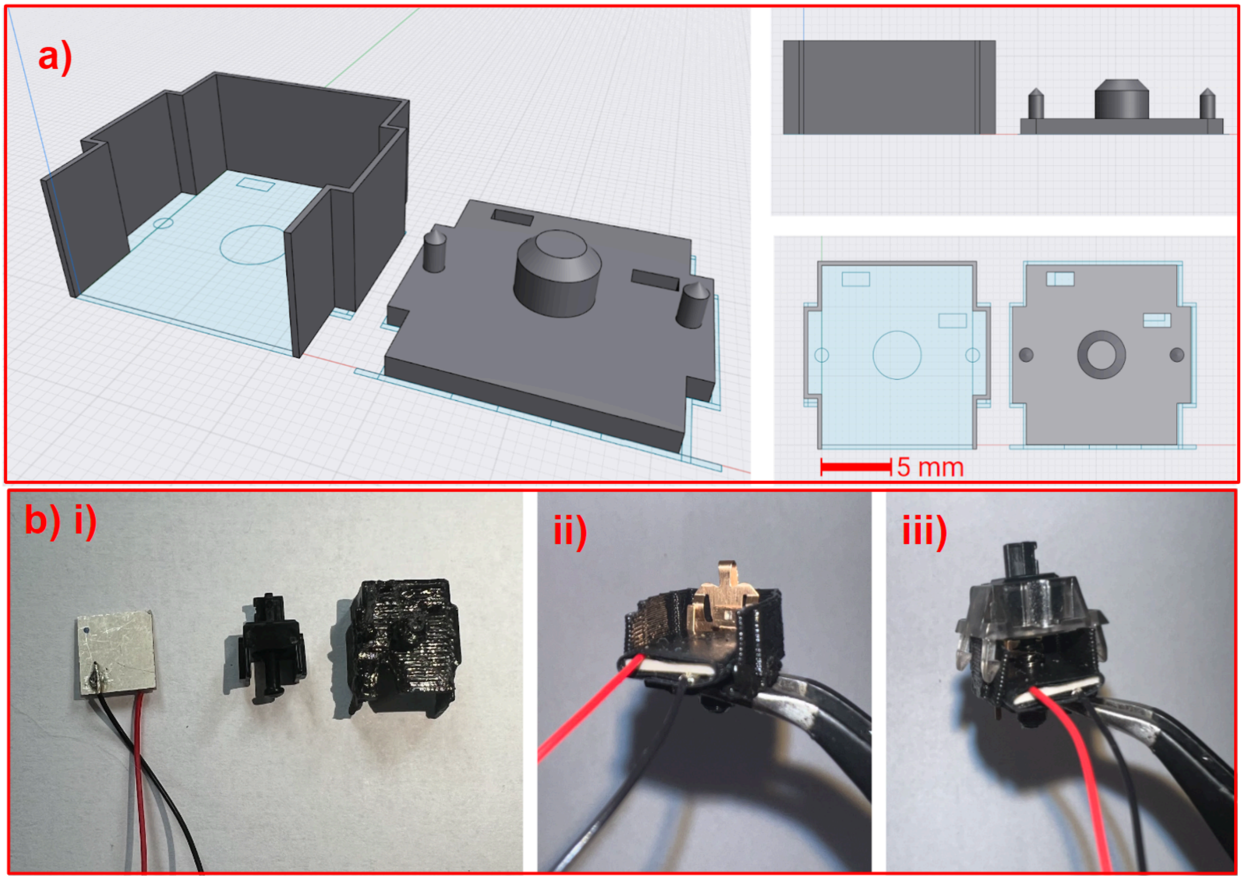
In developing the design for my key prototype, I aimed to maximize the utilization of mechanical energy generated by keystrokes during typing. To achieve this goal, I selected a mechanical key switch as the basis for my key mechanism. In a standard key switch, when pressed, the stem makes contact with the bottom of the housing base. To accommodate the integration of the piezoelectric plate, I reconfigured the housing base of the key switch to match the piezoelectric plate's dimensions. I also removed one wall of the housing base so that the piezoelectric wiring is easily accessible for testing (Figure 6).

In a conventional key switch design, a spring is typically held in place by a tube within the housing base, and the stem exerts pressure on the spring as it travels through the bottom of the housing base. Additionally, I made modifications to shorten the stem, thereby allowing the piezoelectric plate to rest at the base of the key switch, ensuring that pressure is applied from the stem while typing. This adaptation was made to enhance the capture of mechanical energy during typing for energy harvesting purposes.

Following the printing process, the piezoelectric plate was carefully wired with positive and negative leads and integrated into the housing base, along with the inclusion of metal leaves. To ensure the stability of the spring within the assembly, a layer of double-sided electrical tape was applied atop the piezoelectric element, providing secure adherence to the spring
(Figure 6bii). Subsequently, the shortened stem was affixed onto the assembly, and the upper housing was utilized to encase the entire device, thereby finalizing the construction
(Figure 6biii).

Figure 6

Key Design and Assembly

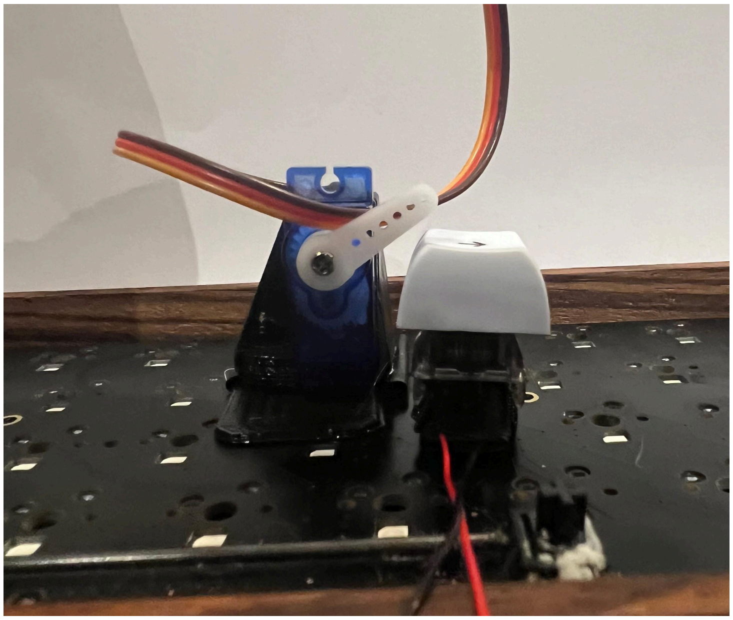


Statistical Analyses

To demonstrate a proof of concept, I wanted to determine how much voltage was being produced by my design for every keystroke. I attached positive and negative leads to the piezoelectric plate and used a voltage probe to graph the results as the key is pressed uniformly by the SG90 motor over a period of time (Figure 7)

Figure 7

Computer key prototype attached to keyboard;



Note: The force applied by SG90 motor results in a voltage

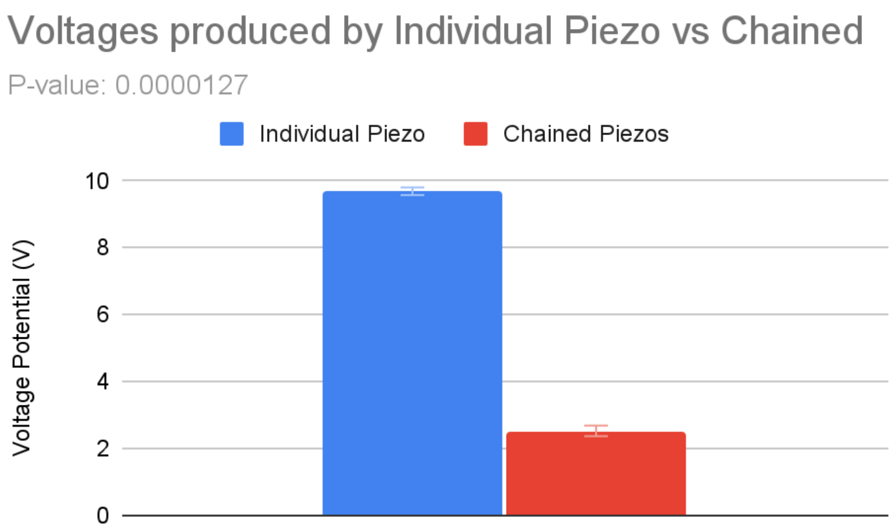
Results

Phase I

The results indicate that the piezoelectric chain beneath the keyboard generated varying outputs depending on the applied pressure. During testing, individual piezo elements produced average voltages ranging from 0.8 to 2.4 volts, with the potential to reach up to 40 volts under increased pressure. However, once the piezo elements were soldered together, the average recorded voltage for a single piezo element within the chain was 2.52 volts, significantly lower than when tested individually. This voltage drop is supported by a statistically significant p-value of 0.00000127. Several factors contribute to this decrease, as discussed in the limitations section. The findings highlight the impact of pressure and soldering on voltage production, emphasizing the importance of assembly methods.

Figure 8

Comparing the voltages produced by applying pressure to chained piezo and individual piezo elements

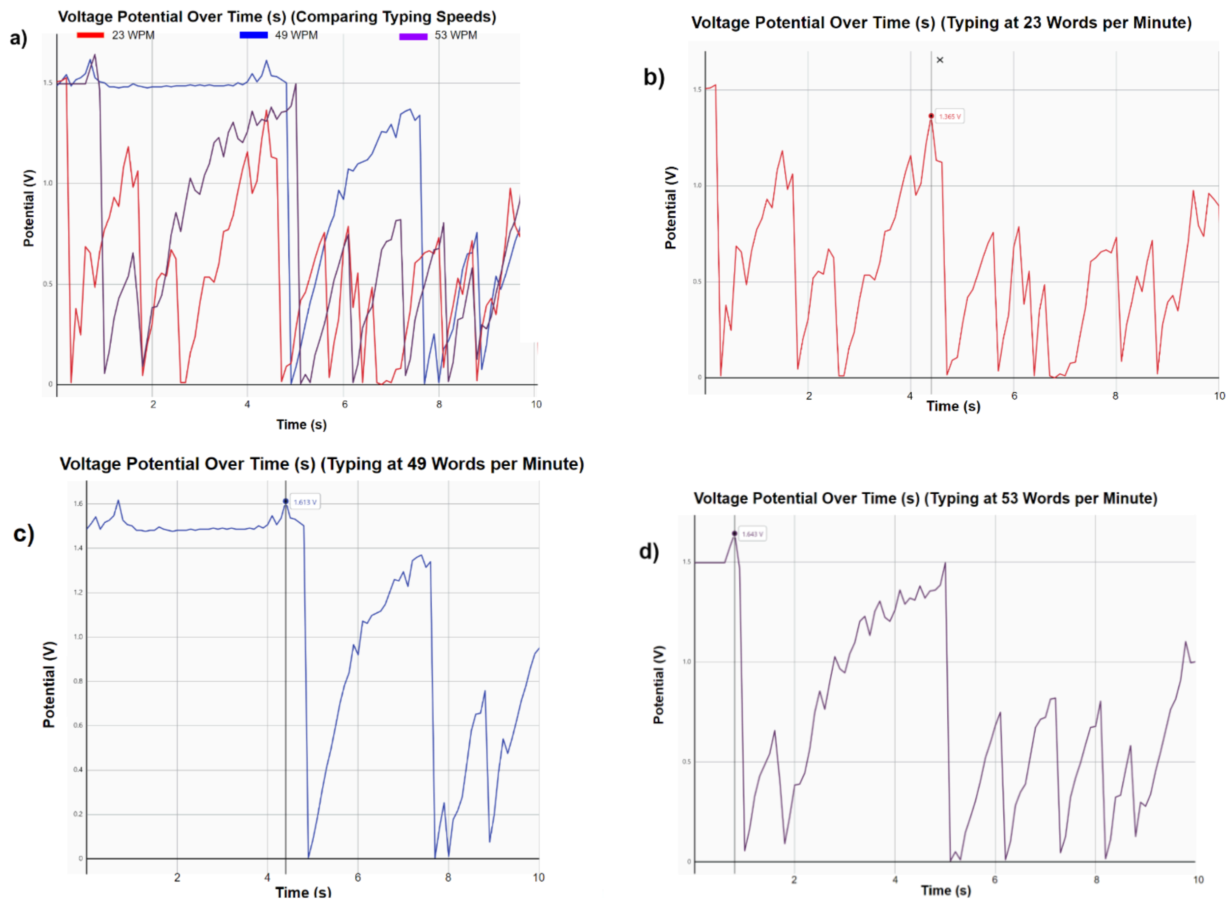


Note: Both the individual and piezo in the chain were repeatedly pressed simultaneously for 10 seconds with approximately the same amount of repeated pressure.

A specified typing test was conducted using an online typing test platform. Three trials were completed, each lasting 10 seconds, to mimic real-life computer usage accurately. The typing speeds in words per minute (wpm) for these trials were recorded as 23 wpm, 49 wpm, and 53 wpm, and the corresponding voltage accumulation was measured. These findings are presented in Figure 9, illustrating the relationship between typing speed and electrical potential collected.

Figure 9

Results of 3 Typing Test Trials



Note: All compare time spent typing to the electrical potential collected **a)** three trials (23, 49, and 53 wpm) on one graph for easier comparison; **b)**23 wpm, Peak: 1.527V, Mean: 0.436V;
c) 49 wpm, Peak: 1.618V, Mean: 1.029 V; **d)** 53 wpm, Peak: 1.643V, Mean: 0.635V.

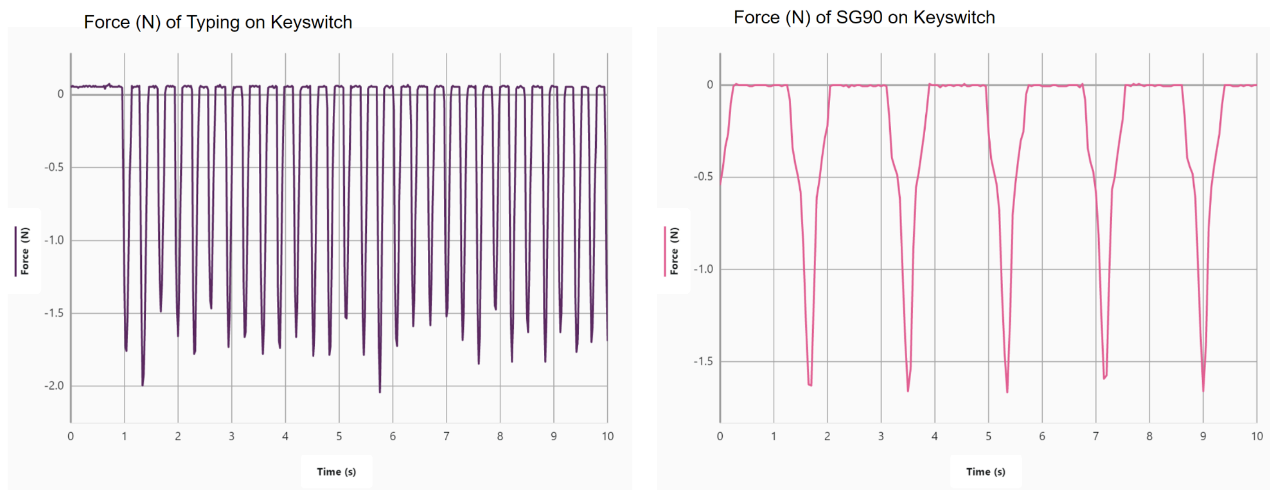
Phase II

Background Testing

The graph in Figure 10 depicts the pressure in Newtons of pressing a key switch. 20 samples were taken per second over 10 seconds. The average peak was -1.679 N. Based on these results, the SG90 was then coded and assembled to strike the key with a force of 1.667 N and was kept constant across all further tests.

Figure 10

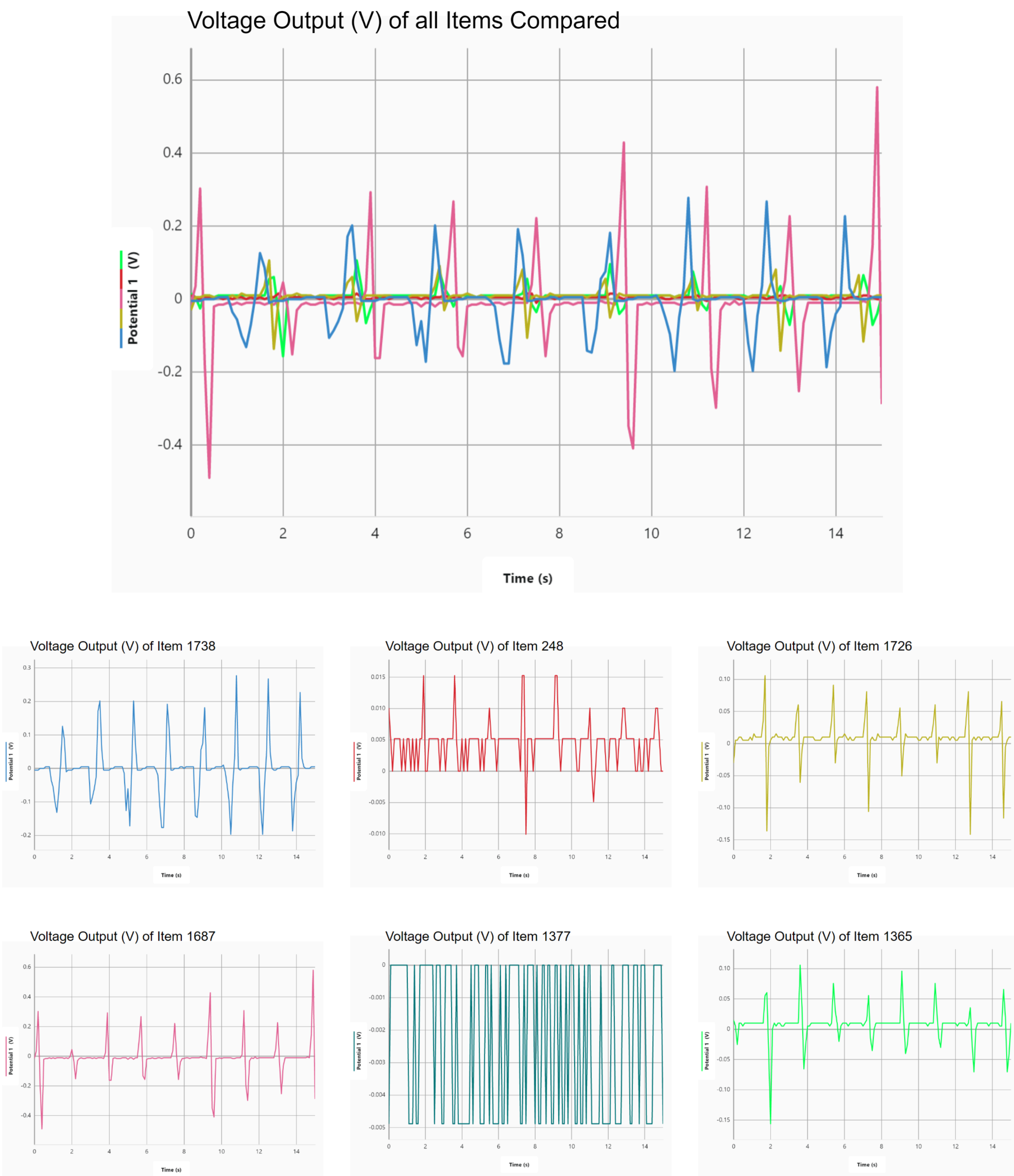
Determination of Typing Force



The graph in Figure 11 shows the voltage graphs of each piezoelectric material applied with 1.667 Newtons of force to best select a material to integrate into the computer key. It is displayed that Item 1687 produced the largest voltage with a range of 0.045V-0.580V, followed by Item 1738 (0.126V-0.277V), Item 1365 (0.060V-0.106V), Item (0.056V-0.106V), Item 248 (0.005V-0.015V). Item 1377 broke under the force of 1.667N and therefore did not read any positive voltage.

Figure 11

Comparing the Voltage Outputs of Different Piezoelectric Ceramics



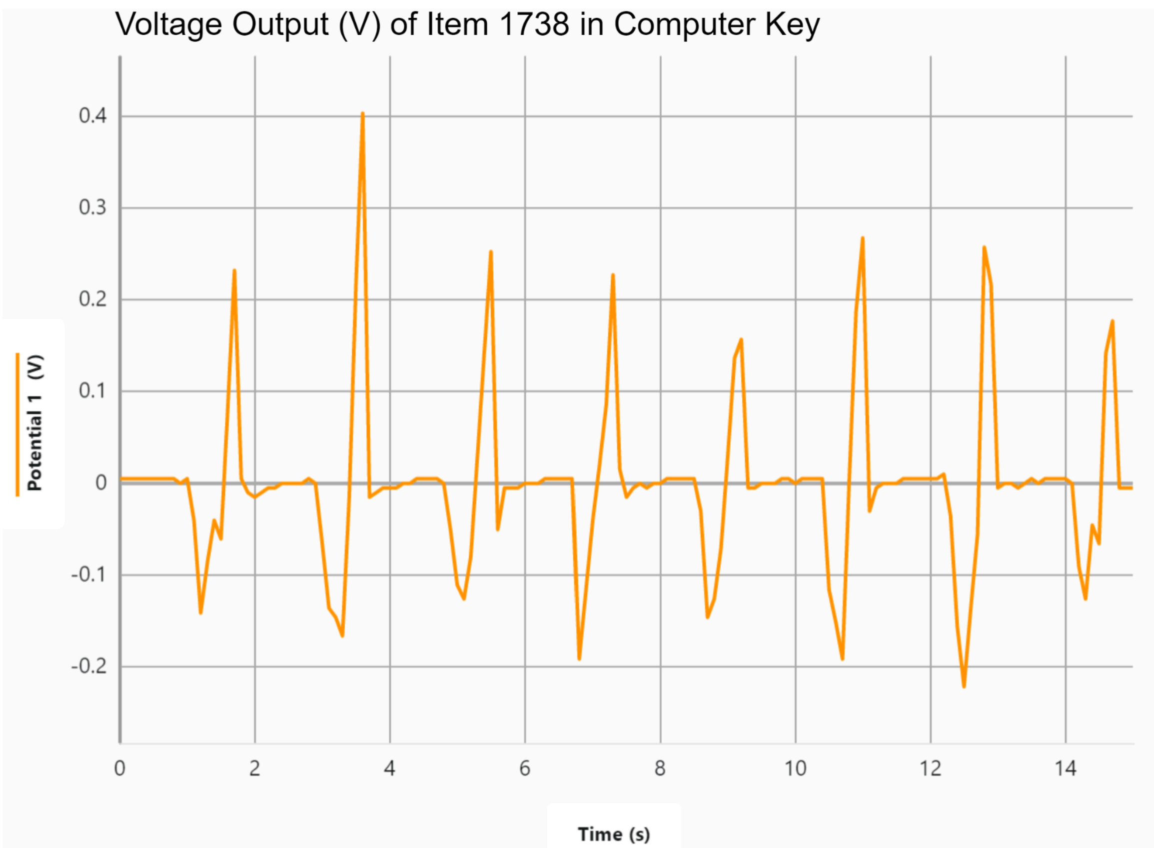
Note*: Item 1377 cracked during testing and therefore did not provide a positive voltage reading.*

Voltage output

To test the ability of the computer to harvest electricity through typing, Item 1738 was incorporated into the engineering computer key, and the SG90 motor applied force to replicate typing over 15 seconds. The maximum voltage reading was 0.403V per keystroke (Figure 12).

Figure 12

Voltage production of Computer Key Mechanism



Discussion

This study is the first to test proof of concept for a self-sustaining computer. Here, I present an analysis of my findings relating to voltage production to determine the feasibility of my design.

Phase I

I independently executed background research reading and annotating over 20 journal articles in the field of energy harvesters and piezoelectric material, as well as had first-hand experience working with piezoelectric energy harvesters. Therefore, objective 1 of exploring piezoelectric energy harvesters was successfully accomplished. However, the assembly process led to a significant decrease in generated voltage when piezo elements were soldered into a chain, suggesting issues with soldering or energy dispersion within the chain. The typing trials revealed that typing speed had less impact on voltage production than the pressure applied to the keys, with faster typing often resulting in less pressure. The design generated electricity but was insufficient to fully charge the laptop, pointing to the need for more efficient piezoelectric materials.

The study was limited by material availability, human error during assembly, and the laptop's age and inefficient battery. Future work includes replicating the design with more efficient sensors and exploring various PZT configurations. Using square PZT directly on the keys to maximize pressure and consulting field experts could yield significant advancements in the design's performance.

Phase II

Phase I of this study was able to demonstrate the potential of harvesting energy from typing using piezoelectric material. I obtained a one-year provisional patent on my research, and presented my findings at the Regeneron: International Science and Engineering Fair where I was awarded 4th Place in the Energy: Sustainable Materials and Design Category. Based on the results and limitations of my first study, I chose to continue this project using higher-quality piezoelectric materials and focus on the creation of a singular, reproducible key.

Figure 11 shows a positive accumulation of voltage for applied pressure similar to that of a keystroke to various piezo elements. My goal in this process was to show proof of concept for a self-charging computer. Though the voltage produced is relatively low, my results suggest that there is potential for increasingly positive results given better quality resources. Therefore, a fully autonomously charging computer that utilizes clean, renewable energy is achievable.

Material Selection

My first objective was to compare the qualities of hard and soft piezoelectric ceramics. Of the materials obtained, Items 248 and 1687 were APC material 840, categorized as a hard piezoelectric ceramic. Items 1738, 1726, 1377, and 1365 are material 850, which are soft piezoelectric materials. Soft materials have a lower elastic modulus, making them more deformable under mechanical stress. This flexibility allows them to generate electrical signals even with smaller mechanical deformations, which I thought would be more valuable in my design which involves small amounts of applied force/mechanical energy. However, soft materials typically have lower piezoelectric constants, resulting in smaller electric charges or voltages generated when mechanically deformed. Therefore, I was unsure if soft materials would be better suited for energy harvesting in computer keys due to these conflicting characteristics, and elected to test the materials on my own.

In our evaluation of voltage outputs from both hard and soft piezoelectric materials, there was no discernible advantage of one material type over the other. Specifically, the hard material exhibited the highest and fourth-highest voltages, while the soft material yielded the second and third-highest voltages as displayed in Figure 11. In the context of my design, material type did not demonstrate a significant performance preference. Instead, the thickness of the material emerged as a more influential factor affecting voltage output.

Notably, the results indicated that thinner materials outperformed thicker ones in converting mechanical energy from typing into electricity, while still possessing the necessary thickness to withstand the pressure exerted by the keys, considering the inherently brittle nature of PZT materials. This became evident in the case of Items 248 and 1365, both of which had a thickness of ≥ 1mm and displayed the lowest recorded voltage outputs. In light of the observation that Item 1377 experienced cracking during testing, I decided to employ Item 1738 in the computer key instead of Item 1687. This choice was informed by Item 1738's favorable combination of high voltage output and greater thickness compared to Item 1687, which reduced the risk of potential fractures during testing.

Piezoelectric feasibility

My second objective aimed to assess whether the mechanical pressure exerted during typing was sufficient to warrant the incorporation of piezoelectric elements. To measure the average force applied to a key during typing, a series of tests utilizing the Vernier Force Probe mechanism revealed an average force of 1.667 N. This force was replicated by the SG90 motor and uniformly applied to all the piezoelectric materials. Remarkably, each material exhibited a positive voltage response when fully intact.

These findings underscore the notion that the force typically applied to keys during typing is indeed substantial enough to trigger a piezoelectric response. As a result, the feasibility of piezoelectric technology as an energy harvesting method for such devices justifies further exploration and consideration.

My third and final objective was to test piezoelectric performance within my key design. As stated above I used Item 1738 in the engineered computer key, and again replicated the process of typing using the SG90 motor. This test was most comparable to the process of typing as the force was applied to the key, in which the stem of the key switch strikes the piezoelectric material. It was recorded that the maximum voltage for a keystroke using 0.6 N of force was 0.403V. This value was higher than the maximum value for Item 1738 outside of the key complex, showing how the process of typing is feasible for piezoelectric energy harvesting. While the figure of 0.403 V may not appear monumental in isolation, it is crucial to recognize that each keystroke has the potential to accumulate energy over extended durations. Therefore, the concept of leveraging piezoelectric materials for device charging emerges as a viable and pragmatic prospect.

This study provides novel insights into the vast application of piezoelectric materials for clean energy generation and is the first study done to show proof of concept for a fully self-charging computer design utilizing piezoelectricity. My research successfully demonstrates the viability of a self-charging computer using piezoelectric transducers. Not only does a piezoelectric keyboard provide greater convenience to laptop users by eliminating annoyances associated with limited battery life, but it also helps to popularize piezoelectricity, a clean and renewable energy source for daily applications.

Piezoelectricity is a novel concept that allows energy to be produced out of what appears to be nothing (Sharma et al. 2019). My current results, combined with studies done regarding piezoelectric roads and the forms of sensors used to harvest large amounts of energy (Wang et al. 2020), support the idea that further research of this design involving stronger piezoelectric transducers will bear the potential to *fully* charge a laptop while typing. It can't be overlooked that the current design still provides stable power to the computer through keystrokes, resulting in longer periods of usage between charging. My study supports prior studies done by Xu et al in 2017, Sun et al in 2019, and Petritz et. al in 2021, which all explain how piezoelectricity is a viable energy harvester that can be applied to numerous aspects of daily life, and continued research in the field for stronger materials causes all piezoelectric designs to continue their climb to maximum efficiency. Nearly all piezoelectric transducers are on the rise to make big impacts in the renewable energy field, specifically related to clean energy generation methods, and have already been discussed in power-generating sidewalks, roads, and even biomedical devices (Akinga et al. 2020, Chaturvedi et al. 2013, Xu et al. 2017, Petritz et al. 2021)**.** This study also supports efforts to look for flexible, thin piezoelectric solutions to yield the greatest energy output.

This research marks a significant milestone in the pursuit of innovative solutions to the world's energy challenges. The concept of utilizing piezoelectric materials to harness mechanical energy for self-sustaining computer applications offers a practical and sustainable approach to reducing our carbon footprint and alleviating the dependence on non-renewable energy sources. As the global community intensifies its commitment to mitigating climate change and adopting cleaner energy practices, the findings of this research contribute to the energy harvesting and piezoelectric field by demonstrating the feasibility of integrating renewable energy solutions into everyday life. Laptops are some of the most widely used electronic devices, and this study calls to recognize how we can reduce carbon emissions in sectors commonly overlooked. The prospect of autonomous, clean energy charging in computers signifies a substantial leap toward a more sustainable future, where cutting-edge technology and environmental preservation coexist harmoniously. The innovative solutions presented in this study invite further exploration and research, opening new possibilities for reducing energy-related carbon emissions and promoting a cleaner, greener world

Limitations and Future Work

The major limitation of this study was the material worked with. Since this research was conducted entirely by the student researcher, there was limited access to the range of piezoelectric materials I could test. American Piezo provided cost-efficient materials to use in my prototype, but further experimentation with new materials is certainly necessary to demonstrate the full capabilities of my design. The brittle ceramics used in this study could be easily replaced with the most recently explored thin, flexible materials explored by Qian and others in March 2023.

Conclusion

This two-year study showcases the potential of piezoelectric materials for self-charging computer designs, a pioneering concept. It demonstrates the viability of a piezoelectric keyboard, enhancing user convenience and promoting clean and renewable energy use in daily life. While the current design provides stable power to the computer through keystrokes, the research suggests that with stronger piezoelectric transducers, fully charging a laptop while typing may become a reality, aligning with previous studies on piezoelectric energy's potential applications. This marks a significant milestone in addressing global energy challenges, offering practical and sustainable ways to reduce carbon emissions and decrease dependence on non-renewable energy sources. It signifies a substantial leap toward a greener future, where cutting-edge technology and environmental preservation coexist seamlessly, inviting further exploration and research in the field.

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